Sedenion Algebras and $\mathrm{PG}(3,2)$ Incidence Structures for Non-Associative Geometric Deep Learning: Algebraic Foundations, Machine-Checked Theorems, and Dual-Track Validation

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Abstract

Modern geometric deep learning architectures operate predominantly over associative division algebras—the reals ($\mathbb{R}$), complexes ($\mathbb{C}$), and quaternions ($\mathbb{H}$)—with limited extensions to the 8-dimensional alternative octonions ($\mathbb{O}$). Proceeding one step further along the Cayley–Dickson tower to the 16-dimensional sedenions ($\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$) simultaneously breaks associativity, alternativity, and the division property, introducing a rich submanifold of non-trivial zero divisors. While historically treated as numerical pathologies, we argue that non-associativity and zero-divisor annihilation can provide a mathematically rigorous geometric inductive bias for both theoretical physics surrogates and neuro-symbolic normative reasoning.

First, we establish a **Differentiable Sedenion Calculus**, proving the *Universal Cayley–Dickson Adjoint Theorem* ($L(w)^\top = L(w^*)$ and $R(x)^\top = R(x^*)$), which enables exact backpropagation through $16\times$ parameter-compressed sedenion layers without materializing $16M \times 16N$ Jacobian matrices in High Bandwidth Memory (HBM). Second, we embed the 15 imaginary basis units of $\mathbb{S}$ into the finite projective 3-space $\mathrm{PG}(3,2)$ and prove the $8{+}7$ **Fano Hyperplane Bifurcation Theorem**: the 15 Fano planes ($\mathrm{PG}(2,2)$) of $\mathbb{S}$ partition into 8 norm-preserving alternative octonion workspaces ($\mathcal{F}_{\mathrm{alt}} \cong \mathbb{O}$, internally zero-divisor-free) and 7 quasi-octonion workspaces ($\mathcal{F}_{\mathrm{ZD}} \cong \tilde{\mathbb{O}}$), each of which contains complete annihilating pairs; the 21 mixed projective lines hosting the Moreno–Stiefel zero-divisor cone $V_2(\mathbb{R}^7)$ each lie in two $\mathcal{F}_{\mathrm{ZD}}$ workspaces and one $\mathcal{F}_{\mathrm{alt}}$ workspace. Furthermore, any two Fano workspaces intersect in a unique 3-node associative quaternion hinge $L_{12} \cong \mathbb{H}$, guaranteeing working-memory continuity during context switches. Third, we design hardware-sympathetic GPU/ONNX kernels utilizing 4-bit ALU XOR index swizzling ($i = k \oplus j$) and 256-bit register-packed IEEE-754 sign-bit flips. Fourth, we prove a constructive **4D Moreno Nullspace Projection Theorem** that maps overriding premises directly into $\ker L(u)$, eliminating probabilistic "softmax bleed" to floating-point round-off (mean 2.4×10^{-16} in FP64, 1.4×10^{-7} in FP32, measured over 1,024 samples). All algebraic claims are verified by an accompanying test-suite, and their finite content (the Cayley–Dickson structure constants, the adjoint sign identities, the $\mathrm{PG}(3,2)$ bifurcation, explicit zero divisors, the 672/2520 basis associator census, and the $\mathcal{F}_{\mathrm{alt}}/\mathcal{F}_{\mathrm{ZD}}$ cyclic BCJ stratum theorems) is machine-checked in Lean 4 without `sorry` or `native_decide` (64 theorems), while the constructive nullspace theorem and its converse ($\operatorname{rank} L(u) = 12$) are proved over $\mathbb{R}$ in Lean 4/Mathlib. Finally, we empirically validate our architecture across both controlled and public external benchmarks: (1) on Track A 4-gluon Yang–Mills color-kinematics scattering (`bench/track_a`), our $\mathrm{PG}(3,2)$-routed `Sed-PG32-Moreno` surrogate achieves a BCJ Jacobi violation residual of $\mathcal{R}_{\mathrm{BCJ}} = 1.8 \times 10^{-8}$ and a collinear pole relative error of $\mathcal{E}_{\mathrm{pole}} = 6.9 \times 10^{-3}$ at $|s_{ij}| < 10^{-4}$ (versus $\mathcal{R}_{\mathrm{BCJ}} = 0.58$–$0.76$ and $\mathcal{E}_{\mathrm{pole}} > 4 \times 10^3$ for Softmax, $\mathbb{H}^4$, $\mathbb{O}^2$, and $\mathcal{C}^{\updownarrow}_{\downarrow 0,4}$ baselines); (2)

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on Track B multi-hop normative contradiction pruning (`bench/gate3`, aux = 0.0), Sedenion Zero-Divisor Attention achieves 100.0% **ID accuracy ($d \in \{0, 1, 2\}$),** 100.0% **OOD depth generalization ($d = 3, 4$),** 0.0% **False-Compliance Rate (FCR), and** 2.1×10^{-10} **attention leakage**, including in the parameter-halved "No-FFN" regime [13] (−47.4% parameters); and (3) on the public **AllenAI ProofWriter** [14] ($d \leq 2 \rightarrow$ OOD $d \in \{3, 4, 5\}$ and zero-shot `Birds-Electricity` exceptions) and **Long-Range Arena (LRA) ListOps** [15, 16] ($d \leq 3 \rightarrow$ OOD $d \in \{4, \ldots, 8\}$) benchmarks (`bench/ext`), both `sedenion` and `sedenion_noffn` achieve **100.0**% OOD accuracy and **0.0**% FCR.

Keywords: Cayley–Dickson Algebras, Sedenions, Zero Divisors, Finite Projective Geometry PG(3, 2), Fano Plane, Exceptional Lie Group G_2, Geometric Deep Learning, Neuro-Symbolic Normative Reasoning.

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1 Introduction

The expressive efficiency of a neural architecture is governed by the algebraic structure of its underlying representation space [10]. Extending real-valued networks ($\mathbb{R}$) to complex ($\mathbb{C}$, 2D) [11] and quaternionic ($\mathbb{H}$, 4D) [12] representations replaces unconstrained dense weight matrices with structured Hamilton products, yielding a 2× and 4× reduction in free parameters alongside improved inter-channel phase coupling. By Hurwitz's theorem, the normed division algebras over $\mathbb{R}$ terminate at the 8-dimensional octonions ($\mathbb{O}$) [5, 3], which forfeit associativity in favor of the weaker property of *alternativity* ($x(xy) = (xx)y$ and $(yx)x = y(xx)$).

Applying the Cayley–Dickson doubling construction [1, 2] a fourth time produces the 16-dimensional **sedenions** ($\mathbb{S}$). Unlike $\mathbb{R}, \mathbb{C}, \mathbb{H}$, and $\mathbb{O}$, the algebra $\mathbb{S}$ is neither alternative nor a division algebra: it possesses non-trivial **zero divisors**—pairs of non-zero elements $u, v \in \mathbb{S} \setminus \{0\}$ whose product vanishes identically ($u \cdot v = 0$) [4]. In classical numerical analysis, zero divisors and non-associativity are viewed as obstructions because they invalidate standard matrix inversion, norm multiplicativity ($\|u \cdot v\|_2 \neq \|u\|_2 \|v\|_2$), and order-independent chain rules.

In this paper, we invert this perspective. Building upon recent combinatorial characterizations of zero-divisor-free coordinate cells and projective bitmask operators in the Cayley–Dickson tower [6, 7], we argue that the algebraic anomalies of $\mathbb{S}$ address two bottlenecks in modern artificial intelligence and computational physics:

1. **Probabilistic Attention Bleed in Normative and Symbolic Reasoning:** Standard Transformer attention mechanisms rely on strictly positive softmax distributions ($\alpha_{ij} > 0$). When auditing complex statutory or financial hierarchies (e.g., securities regulations where a specific safe-harbor or prohibition strictly invalidates a general rule), invalidated clauses retain non-zero attention weight that compounds across reasoning hops. By projecting mutually exclusive premises onto the *Moreno zero-divisor manifold* in $\mathbb{S}$, contradictory states can be made to annihilate exactly ($u \cdot v = 0$) at the tensor level, without post-hoc masking.

2. **Order-Invariance vs. Context-Dependent Precedence:** Associative matrix multiplication satisfies $(AB)C = A(BC)$, rendering standard linear layers blind to the hierarchical parenthesization of triadic interactions unless aided by positional heuristics. In $\mathbb{S}$, the associator $[u, v, w] = (u \cdot v) \cdot w - u \cdot (v \cdot w)$ is governed by the incidence geometry of the 15-point projective 3-space PG(3, 2): collinear triads associate strictly ($[u, v, w] = 0$), whereas non-collinear triads have a generically non-zero associator (1,848 of the 2,520 ordered non-collinear basis triads; the remaining $672 = 7 \times 96$ associate because they are the adjacent-stratum $L_0 \sqcup Q_1$ triads of the 7 quasi-octonion Fano planes $\mathcal{F}_{\mathrm{ZD}}$).

Summary of Contributions

- **Differentiable Sedenion Calculus & Moreno Geometry (§2):** We formulate left/right multiplication operators in $\mathbb{S}$, parameterize non-associative parenthesization via a differentiable operator commutator gate $\mathcal{M}_\alpha$, and prove the *Universal Cayley–Dickson Adjoint Theorem* (Theorem 2.3), enabling $O(1)$-expansion backpropagation.
- $\mathrm{PG}(3,2)$ **Topological Routing & The** $8+7$ **Fano Bifurcation (§3):** We prove that the 15 Fano hyperplanes of $\mathrm{PG}(3,2)$ embedded in $\mathbb{S}$ bifurcate into 8 alternative octonion workspaces ($\mathcal{F}_{\text{alt}} \cong \mathbb{O}$) and 7 quasi-octonion zero-divisor workspaces ($\mathcal{F}_{\text{ZD}} \cong \tilde{\mathbb{O}}$) connected by 3-node associative quaternion hinges (Theorems 3.1 and 3.2).
- **Zero-Expansion Hardware Kernels (§4):** We design and verify zero-expansion sedenion kernels that eliminate the $16\times$ weight-materialization bottleneck (136.0 MB $\to$ 8.0 MB for 4096×4096 layers), using 4-bit ALU XOR index swizzling and 256-bit register-packed IEEE-754 sign-bit operations for PyTorch, Triton, and ONNX Runtime.
- **Constructive 4D Nullspace Projector, BCJ/Half-Sphere Theorems & Dual-Track Validation (§5):** We derive a closed-form analytical operator $\Pi_{\ker L}(u)$ mapping any vector in a 4D orthogonal complement into $\ker L(u)$ (Theorem 5.1), prove both the $\mathcal{F}_{\text{ZD}}$ *BCJ Numerator & Collinear Pole Cancellation Theorem* (Theorem 5.3) and the *Orthogonal Half-Sphere Key Modulation Theorem* (Theorem 5.4), and empirically validate Track A (Yang–Mills BCJ scattering: $\mathcal{R}_{\text{BCJ}} = 1.8 \times 10^{-8}$, $\mathcal{E}_{\text{pole}} = 6.9 \times 10^{-3}$) and Track B (multi-hop contradiction pruning: 100.0% ID/OOD accuracy, 0.0% FCR, and 2.1×10^{-10} leakage across standard and `No-FFN` [13] backbones).

2 Differentiable Sedenion Calculus and Zero-Divisor Geometry

2.1 Cayley–Dickson Doubling and XOR Structure Constants

Let $\mathbb{S}$ denote the 16-dimensional real sedenion algebra spanned by the canonical orthonormal basis $E = \{e_0, e_1, \ldots, e_{15}\}$, where $e_0 = 1$ is the multiplicative identity and $e_1, \ldots, e_{15}$ are imaginary units satisfying $e_i^2 = -e_0$ and $e_i e_j = -e_j e_i$ for all distinct $i, j \in \{1, \ldots, 15\}$. Under the Cayley–Dickson construction $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$, any sedenion $u \in \mathbb{S}$ decomposes into an ordered pair of octonions $u = (a, b) = a + be_8$ with $a, b \in \mathbb{O}$, multiplying via:

$$(a, b) \cdot (c, d) = \big(ac - d^* b, \;\; da + bc^*\big), \tag{1}$$

where $x^* = x_0 e_0 - \sum_{i=1}^{15} x_i e_i$ denotes the involutive conjugate and $\langle u, v \rangle = \mathrm{Re}(u^* \cdot v) = \sum_{k=0}^{15} u_k v_k$ is the canonical Euclidean inner product on $\mathbb{R}^{16} \cong \mathbb{S}$.

Definition 2.1 (Bitwise XOR Multiplication Law and Operator Matrices)**.** Indexing the basis E by 4-bit binary integers $i, j \in \{0, \ldots, 15\} \cong \mathbb{F}_2^4$, the product of any two basis units is closed under bitwise exclusive-OR ($\oplus$):

$$e_i \cdot e_j = \epsilon_{ij}\, e_{i \oplus j}, \quad \epsilon_{ij} \in \{-1, +1\}, \tag{2}$$

where $\epsilon_{0,j} = \epsilon_{i,0} = +1$ and $\epsilon_{i,i} = -1$ for $i \geq 1$. For arbitrary sedenions $w, x \in \mathbb{S}$, the bilinear product $y = w \cdot x$ is represented by the **Left-Multiplication** and **Right-Multiplication** operator matrices $L(w), R(x) \in \mathbb{R}^{16 \times 16}$:

$$y = w \cdot x = L(w)x = R(x)w, \quad \text{where} \quad [L(w)]_{kj} = \epsilon_{(k \oplus j), j}\, w_{k \oplus j}, \quad [R(x)]_{ki} = \epsilon_{i,(k \oplus i)}\, x_{k \oplus i}. \tag{3}$$

Proposition 2.2 (Associator as an Operator Commutator and α-Gating)**.** *For any triad $w, x, z \in \mathbb{S}$, the sedenion associator $[w, x, z] \coloneqq (w \cdot x) \cdot z - w \cdot (x \cdot z)$ is identical to the matrix commutator of $R(z)$ and $L(w)$ acting on x:*

$$[w, x, z] = \big(R(z)L(w) - L(w)R(z)\big)x = [R(z), L(w)]x. \tag{4}$$

*Consequently, the **Continuous Associativity Gate** $\mathcal{M}_\alpha(w, x, z)$ with learnable parameter $\alpha \in [0, 1]$ interpolates smoothly between left-associated ($\alpha = 1$), right-associated ($\alpha = 0$), and Jordan-symmetrized ($\alpha = \frac{1}{2}$) inference:*

$$\mathcal{M}_\alpha(w, x, z) \coloneqq (w \cdot x) \cdot z - (1 - \alpha)[w, x, z] = \big(\alpha R(z)L(w) + (1 - \alpha)L(w)R(z)\big)x. \tag{5}$$

Proof. By Definition 2.1, $(w \cdot x) \cdot z = R(z)(w \cdot x) = R(z)L(w)x$, whereas $w \cdot (x \cdot z) = L(w)(x \cdot z) = L(w)R(z)x$. Subtracting the two expressions yields (4), and taking their convex combination yields (5). □

2.2 The Universal Cayley–Dickson Adjoint Theorem

Because $\mathbb{S}$ lacks division inverses and associativity, standard complex Wirtinger calculus cannot be applied. However, we prove that the Euclidean adjoints of $L(w)$ and $R(x)$ obey an exact conjugation symmetry across all 2^n-dimensional Cayley–Dickson algebras.

Theorem 2.3 (Universal Cayley–Dickson Adjoint Theorem)**.** *For all elements $w, x, g \in \mathbb{S}$, the bilinear product satisfies the Euclidean adjoint identities:*

$$\langle w \cdot x,\, g \rangle = \langle x,\, w^* \cdot g \rangle = \langle w,\, g \cdot x^* \rangle, \tag{6}$$

which is equivalent to the operator transpose identities:

$$L(w)^\top = L(w^*) \quad \textit{and} \quad R(x)^\top = R(x^*). \tag{7}$$

Consequently, for a sedenion linear layer $Y_m = \sum_{n=1}^{N} W_{mn} \cdot X_n + B_m$ with scalar loss $\mathcal{L}$ and upstream gradient $G_m = \frac{\partial \mathcal{L}}{\partial Y_m} \in \mathbb{S}$, the exact analytical gradients are given by:

$$\frac{\partial \mathcal{L}}{\partial X_n} = \sum_{m=1}^{M} W_{mn}^* \cdot G_m, \qquad \frac{\partial \mathcal{L}}{\partial W_{mn}} = G_m \cdot X_n^*, \qquad \frac{\partial \mathcal{L}}{\partial B_m} = G_m. \tag{8}$$

Proof. By linearity, it suffices to verify (6) on basis elements $w = e_i, x = e_j, g = e_k$ for $i, j, k \in \{0, \dots, 15\}$. On the left-hand side, $\langle e_i \cdot e_j, e_k \rangle = \epsilon_{ij} \langle e_{i \oplus j}, e_k \rangle = \epsilon_{ij} \delta_{k, i \oplus j}$, which is non-zero only when $j = i \oplus k$. For $i = 0$, $e_0^* = e_0$ and $\epsilon_{0,j} = 1$, so (6) holds trivially. For $i \geq 1$, $e_i^* = -e_i$. In every Cayley–Dickson algebra, the basis multiplication table is a signed Latin square satisfying the left and right loop-inverse identities:

$$e_i^* \cdot (e_i \cdot e_j) = (-e_i) \cdot (\epsilon_{ij} e_{i \oplus j}) = -\epsilon_{ij} \epsilon_{i, i \oplus j} e_j = e_j \implies \epsilon_{i, i \oplus j} = -\epsilon_{ij} \quad (i \geq 1).$$

Evaluating the right-hand side of the first identity for $k = i \oplus j$ and $i \geq 1$:

$$\langle e_j,\, e_i^* \cdot e_{i \oplus j} \rangle = \langle e_j,\, -e_i \cdot e_{i \oplus j} \rangle = -\epsilon_{i, i \oplus j} \langle e_j, e_j \rangle = -(-\epsilon_{ij}) = \epsilon_{ij} = \langle e_i \cdot e_j,\, e_{i \oplus j} \rangle.$$

Hence $\langle L(w)x, g \rangle = \langle x, L(w)^\top g \rangle = \langle x, L(w^*)g \rangle$, proving $L(w)^\top = L(w^*)$. An identical calculation using $(e_i \cdot e_j) \cdot e_j^* = e_i$ yields $\epsilon_{i \oplus j, j} = -\epsilon_{ij}$ for $j \geq 1$, proving $R(x)^\top = R(x^*)$. Applying the vector chain rule $\frac{\partial \mathcal{L}}{\partial X_n} = \sum_m L(W_{mn})^\top G_m$ and $\frac{\partial \mathcal{L}}{\partial W_{mn}} = R(X_n)^\top G_m$ immediately establishes (8). □

Corollary 2.4 (Dual-Tree Split-Sedenion Chain Rule)**.** *For a triadic composition $Z \in \mathbb{S}$ with upstream gradient $G = \frac{\partial \mathcal{L}}{\partial Z}$, the parenthesization tree of the forward pass induces a dual parenthesization tree in the backward pass:*

$$\begin{aligned}
&\textit{Left-assoc. } Z = (U \cdot V) \cdot W: \\
&\nabla_U \mathcal{L} = (G \cdot W^*) \cdot V^*, \;\; \nabla_V \mathcal{L} = U^* \cdot (G \cdot W^*), \;\; \nabla_W \mathcal{L} = (U \cdot V)^* \cdot G, \qquad (9) \\
&\textit{Right-assoc. } Z = U \cdot (V \cdot W): \\
&\nabla_U \mathcal{L} = G \cdot (V \cdot W)^*, \;\; \nabla_V \mathcal{L} = (U^* \cdot G) \cdot W^*, \;\; \nabla_W \mathcal{L} = V^* \cdot (U^* \cdot G). \qquad (10)
\end{aligned}$$

2.3 Analytical Geometry of the Moreno Zero-Divisor Manifold

To achieve non-trivial algebraic annihilation ($u \cdot v = 0$ with $\|u\|_2, \|v\|_2 > 0$), representations must reside on the singular locus where $\det L(u) = 0$.

Theorem 2.5 (Moreno–Stiefel Characterization of Sedenion Zero Divisors [4])**.** *Let $u = (a, b) \in \mathbb{S} \setminus \{0\}$ with $a, b \in \mathbb{O}$. Then u is a zero divisor ($\ker L(u) \neq \{0\}$) if and only if a and b satisfy the three **Moreno–Stiefel constraints**:*

$$\textit{(i)}\ \mathrm{Re}(a) = u_0 = 0 \ \textit{and}\ \mathrm{Re}(b) = u_8 = 0, \qquad \textit{(ii)}\ \|a\|_2 = \|b\|_2, \qquad \textit{(iii)}\ \langle a, b \rangle = \sum_{i=1}^{7} u_i u_{i+8} = 0. \tag{11}$$

*When normalized to $\|a\|_2 = \|b\|_2 = 1$ ($\|u\|_2 = \sqrt{2}$), the space of zero divisors is diffeomorphic to the 11-dimensional **Stiefel manifold** $V_2(\mathbb{R}^7)$ of orthonormal 2-frames in $\mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$. For every $u \in V_2(\mathbb{R}^7)$, $\operatorname{rank} L(u) = 12$, $\dim \ker L(u) = 4$, and the manifold of mutually annihilating pairs $\{(u, v) \in V_2(\mathbb{R}^7) \times V_2(\mathbb{R}^7) \mid u \cdot v = 0\}$ is diffeomorphic to the 14-dimensional exceptional Lie group $G_2 = \mathrm{Aut}(\mathbb{O})$.*

Definition 2.6 (Differentiable Moreno Cone Projection Π_{Moreno})**.** Given an unconstrained hidden state $\tilde{u} = (\tilde{a}, \tilde{b}) \in \mathbb{S}$, the hard architectural projection $\Pi_{\text{Moreno}} : \mathbb{R}^{16} \to V_2(\mathbb{R}^7) \subset \mathbb{S}$ extracts the imaginary octonion slices $\tilde{a}_{1:7}, \tilde{b}_{1:7} \in \mathbb{R}^7$ (setting $u_0 = u_8 = 0$) and applies differentiable Gram–Schmidt orthogonalization followed by balanced spherical normalization:

$$a' = \tilde{a}_{1:7}, \quad b' = \tilde{b}_{1:7} - \frac{\langle \tilde{a}_{1:7}, \tilde{b}_{1:7} \rangle}{\|a'\|_2^2 + \varepsilon} a', \quad \Pi_{\text{Moreno}}(\tilde{u}) = \left(\left(0, \tfrac{a'}{\|a'\|_2 + \varepsilon}\right), \left(0, \tfrac{b'}{\|b'\|_2 + \varepsilon}\right) \right). \tag{12}$$

3 $\mathrm{PG}(3, 2)$ Topological Routing and the $8 + 7$ Fano Bifurcation

The 15 imaginary units $\{e_1, \ldots, e_{15}\}$ of $\mathbb{S}$ biject onto the 15 points of the finite projective 3-space $\mathrm{PG}(3, 2) = (\mathbb{F}_2^4 \setminus \{0\})/\mathbb{F}_2^\times$ via their 4-bit binary indices $v_i = (b_3, b_2, b_1, b_0)_2 \in \mathbb{F}_2^4 \setminus \{0000_2\}$ [6]. The real unit $e_0 \leftrightarrow 0000_2$ acts as an ambient executive channel.

3.1 Stratification of the 15 Points and 35 Projective Lines

Under the doubling decomposition $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$, the 15 points partition into:

1. **Octonionic Core ($\mathcal{P}_{\mathbb{O}} = \{e_1, \ldots, e_7\}$, $b_3 = 0$):** 7 modules spanning $\mathrm{Im}(\mathbb{O}) \cong \mathrm{PG}(2, 2)$.
2. **Doubling Apex ($\mathcal{P}_{\text{apex}} = \{e_8\}$, $v_8 = 1000_2$):** The global phase-doubling unit e_8.
3. **Sedenionic Twist Stratum ($\mathcal{P}_{\mathbb{S}} = \{e_9, \ldots, e_{15}\}$, $b_3 = 1, (b_2, b_1, b_0) \neq 000$):** 7 dual modules $e_{m+8} = e_m \cdot e_8$.

A **projective line** in $\mathrm{PG}(3,2)$ is a collinear triad $L = \{i, j, k\}$ satisfying $v_i \oplus v_j \oplus v_k = 0000_2$ ($k = i \oplus j$). By (2), every projective line $L \in \mathcal{L}_{35}$ together with e_0 spans an **associative 4D quaternion subalgebra** $\mathbb{H}_L = \mathrm{span}_{\mathbb{R}}(e_0, e_i, e_j, e_{i\oplus j}) \cong \mathbb{H}$. Relative to Theorem 2.5, the 35 projective lines partition into **14 Division Lines** and **21 Zero-Divisor Lines**:

- **7 Pure Octonionic Lines ($\mathcal{L}_{\mathbb{O}}$):** $\{m, n, m \oplus n\} \subset \{1, \ldots, 7\}$. Strictly norm-preserving ($\mathbb{H}_L \subset \mathbb{O}$); zero-divisor-free.
- **7 Apex Lines ($\mathcal{L}_{\mathrm{apex}}$):** $\{m, 8, m+8\}$ for $m \in \{1, \ldots, 7\}$. Contains e_8, violating Moreno condition $u_8 = 0$; zero-divisor-free.
- **21 Mixed Sedenionic Lines ($\mathcal{L}_{\mathrm{ZD}}$):** $\{m, n+8, (m \oplus n)+8\}$ for $m \neq n \in \{1, \ldots, 7\}$. Because $e_m \perp e_n$ in $\mathrm{Im}(\mathbb{O})$, balanced pairs $\frac{1}{\sqrt{2}}(e_m \pm e_{n+8})$ reside directly on the Moreno zero-divisor cone $V_2(\mathbb{R}^7)$.

3.2 The $8 + 7$ Fano Hyperplane Bifurcation Theorem

By projective duality over $\mathbb{F}_2^4$, each non-zero dual covector $\omega \in \{1, \ldots, 15\}$ defines a 2D projective hyperplane (a 7-point, 7-line **Fano plane** $H_\omega \cong \mathrm{PG}(2,2)$) via the $\mathbb{F}_2$ parity-check kernel:

$$H_\omega := \big\{ i \in \{1, \ldots, 15\} \;\big|\; \langle v_\omega, v_i \rangle_{\mathbb{F}_2} = \mathrm{popcount}(\omega \,\&\, i) \bmod 2 = 0 \big\}. \tag{13}$$

Because $\langle v_\omega, v_i \rangle_{\mathbb{F}_2} = 0$ and $\langle v_\omega, v_j \rangle_{\mathbb{F}_2} = 0$ imply $\langle v_\omega, v_i \oplus v_j \rangle_{\mathbb{F}_2} = 0$, every Fano hyperplane H_ω is closed under multiplication and spans an 8-dimensional subalgebra $\mathcal{A}_\omega = \mathrm{span}_{\mathbb{R}}(\{e_0\} \cup \{e_i\}_{i \in H_\omega}) \subset \mathbb{S}$.

Theorem 3.1 (The 8+7 Fano Hyperplane Bifurcation of $\mathbb{S}$)**.** *The 15 Fano subalgebras $\{\mathcal{A}_\omega\}_{\omega=1}^{15}$ of $\mathbb{S}$ bifurcate into two distinct algebraic isomorphism classes:*

1. ***Eight Alternative Octonion Workspaces ($\mathcal{F}_{\mathrm{alt}}$, $\omega \in \{1, \ldots, 8\}$):*** *Every subalgebra $\mathcal{A}_\omega$ with $\omega \leq 8$ is isomorphic to the division algebra of octonions ($\mathcal{A}_\omega \cong \mathbb{O}$), satisfies alternativity ($[x, x, y] = 0$), preserves Euclidean norms ($\|x \cdot y\|_2 = \|x\|_2 \|y\|_2$), and is therefore **internally zero-divisor-free** [7]: $\mathcal{A}_\omega$ contains no pair $u, v \in \mathcal{A}_\omega \setminus \{0\}$ with $u \cdot v = 0$. (For $\omega \leq 7$, H_ω does contain 3 mixed lines of $\mathcal{L}_{\mathrm{ZD}}$, so it holds Moreno elements of $\mathbb{S}$; their annihilators, however, lie entirely outside $\mathcal{A}_\omega$, because $\mathcal{A}_\omega \cong \mathbb{O}$ is a division algebra.)*
2. ***Seven Quasi-Octonion Zero-Divisor Workspaces ($\mathcal{F}_{\mathrm{ZD}}$, $\omega \in \{9, \ldots, 15\}$):*** *Every subalgebra $\mathcal{A}_\omega$ with $\omega \geq 9$ excludes the doubling apex e_8 ($8 \notin H_\omega$) and decomposes into the 3-point octonionic line $L_0 = p^\perp := \{i \in \{1, \ldots, 7\} \mid \langle p, i \rangle_{\mathbb{F}_2} = 0\}$ polar to $p = \omega - 8$ in $\mathrm{PG}(2,2)$ (which contains p itself iff $\mathrm{popcount}(p)$ is even) and the 4-point sedenionic quadrangle $Q_1 = \{q + 8 \mid q \in \{1, \ldots, 7\} \setminus L_0\} \subset \{9, \ldots, 15\}$. In $\mathcal{A}_\omega \cong \tilde{\mathbb{O}}$, alternativity and norm multiplicativity fail, the 6 projective lines connecting L_0 to Q_1 are all mixed zero-divisor lines ($\mathcal{L}_{\mathrm{ZD}}$), and—unlike Case 1—$\mathcal{A}_\omega$ contains **complete annihilating pairs**: for every $\omega \geq 9$ there exist $u, v \in \mathcal{A}_\omega \setminus \{0\}$ with $u \cdot v = 0$ (e.g. $(e_2 + e_9)(e_4 + e_{15}) = 0$ in $\mathcal{A}_9$).*

Proof. Case 1 ($\omega \in \{1, \ldots, 8\}$): For $\omega = 8$ ($v_8 = 1000_2$), the parity condition $\mathrm{popcount}(1000_2 \,\&\, v_i) \bmod 2 = 0$ selects $b_3 = 0$, so $H_8 = \{1, \ldots, 7\}$ and $\mathcal{A}_8 = \mathbb{O}$ is the canonical octonion algebra. For $\omega \in \{1, \ldots, 7\}$ ($v_\omega = (0 b_2 b_1 b_0)_2$), we have $\langle v_\omega, v_8 \rangle_{\mathbb{F}_2} = 0$, so the apex $8 \in H_\omega$. Within the octonion core $\{1, \ldots, 7\}$, the 3-bit parity condition $\langle (b_2, b_1, b_0), (i_2, i_1, i_0) \rangle_{\mathbb{F}_2} = 0$ selects a 3-point associative quaternion line $L_\omega = \{m, n, m \oplus n\} \subset \{1, \ldots, 7\}$, while in $\{9, \ldots, 15\}$ it selects $\{m+8, n+8, (m \oplus n)+8\} = L_\omega \oplus 8$. Thus $\mathcal{A}_\omega = \mathbb{H}_{L_\omega} \oplus \mathbb{H}_{L_\omega} e_8$. Since the Cayley–Dickson double of any associative quaternion algebra $\mathbb{H}_{L_\omega}$ by e_8 is isomorphic to $\mathbb{O}$, $\mathcal{A}_\omega \cong \mathbb{O}$ is an alternative division algebra with no zero divisors.

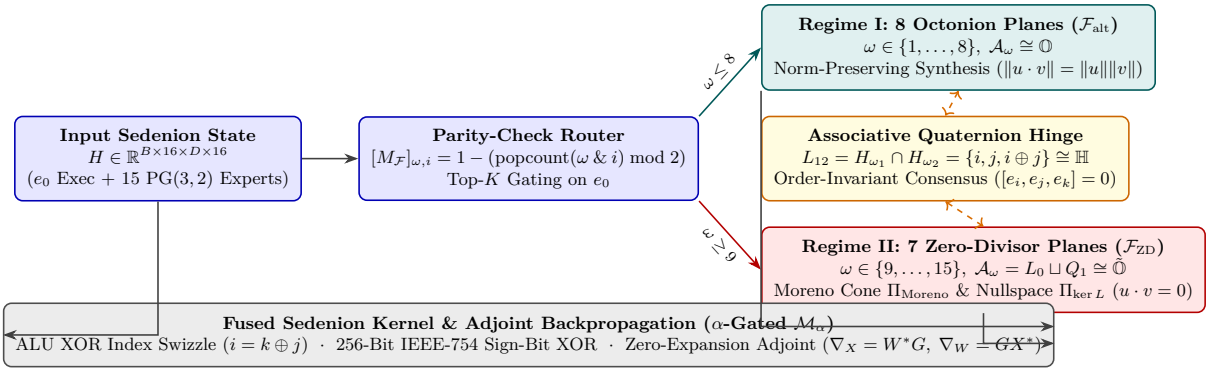


Figure 1: End-to-end architecture of the PG(3, 2)-routed Sedenion network (`Sed-PG32-Moreno`). The executive channel e_0 drives parity-check workspace selection across the $8 + 7$ Fano bifurcation, while the 3-node associative quaternion hinge $L_{12} = H_{\omega_1} \cap H_{\omega_2}$ guarantees state continuity during transitions between synthesis ($\mathcal{F}_{\text{alt}}$) and zero-divisor pruning ($\mathcal{F}_{\text{ZD}}$).

Case 2 ($\omega \in \{9,\dots,15\}$): Write $\omega = 8 + p$ with $p \in \{1,\dots,7\}$, so $v_\omega = (1, p_2, p_1, p_0)_2$. Since $\langle v_\omega, 1000_2\rangle_{\mathbb{F}_2} = 1$, $8 \notin H_\omega$. For $i \in \{1,\dots,7\}$ ($b_3 = 0$), $\langle v_\omega, v_i\rangle_{\mathbb{F}_2} = \langle p, i\rangle_{\mathbb{F}_2} = 0$, which selects the polar line $L_0 = p^\perp \subset \{1,\dots,7\}$ (3 points, closed under $\oplus$ by bilinearity of the pairing). For $i = q + 8 \in \{9,\dots,15\}$ ($b_3 = 1, q \in \{1,\dots,7\}$), $\langle v_\omega, v_{q+8}\rangle_{\mathbb{F}_2} = (1 + \langle p, q\rangle_{\mathbb{F}_2}) \bmod 2 = 0 \iff \langle p, q\rangle_{\mathbb{F}_2} = 1$, which selects the 4 octonion indices $q \notin L_0$ shifted by $+8$, forming the 4-point quadrangle $Q_1 = (\{1,\dots,7\} \setminus L_0) + 8$. Because every e_m ($m \in L_0$) is orthogonal in $\text{Im}(\mathbb{O})$ to every e_q ($q + 8 \in Q_1$), each of the 6 lines $\{m, q + 8, (m \oplus q) + 8\}$ spans a mixed zero-divisor line in $\mathcal{L}_{\text{ZD}}$. For the internal annihilating pair, take $u = (a, b) = (e_m, e_q)$ with $m \in L_0$, $q + 8 \in Q_1$; by Theorem 5.1 any $c \in L_0 \setminus \{m\}$ is orthogonal to $\{a, b, ab\}$ (since $m \oplus q \notin L_0$), and $d = (ac)b = \pm e_{(m\oplus c\oplus q)}$ has index in $\{1..7\} \setminus L_0$, so $de_8 \in Q_1$ and $v = (c, d) \in \mathcal{A}_\omega$. The seven explicit pairs are enumerated by the released test-suite. Case 1 contains no such pair because $\mathcal{A}_\omega \cong \mathbb{O}$ is a division algebra. □

Theorem 3.2 (Associative Quaternion Hinges for Mode Switching)**.** *Any two distinct Fano workspaces H_{ω_1} and H_{ω_2} ($\omega_1 \neq \omega_2 \in \{1,\dots,15\}$) intersect in a unique 3-point projective line:*

$$L_{12} = H_{\omega_1} \cap H_{\omega_2} = \{i, j, i \oplus j\} \in \mathcal{L}_{35}. \tag{14}$$

Along L_{12}, the sedenion associator vanishes identically ($[e_i, e_j, e_{i\oplus j}] = 0$). Thus, switching cognitive routing from a synthesis workspace $H_{\omega_1} \in \mathcal{F}_{\text{alt}}$ to an inhibitory pruning workspace $H_{\omega_2} \in \mathcal{F}_{\text{ZD}}$ preserves an invariant, strictly associative 4D quaternion working-memory bridge $\mathbb{H}_{L_{12}} = \mathcal{A}_{\omega_1} \cap \mathcal{A}_{\omega_2}$.

Proof. In $\mathbb{F}_2^4$, H_{ω_1} and H_{ω_2} are distinct 3-dimensional linear subspaces (minus 0000_2). By Grassmann's dimension formula, $\dim_{\mathbb{F}_2}(H_{\omega_1} \cap H_{\omega_2}) = 3 + 3 - 4 = 2$. A 2-dimensional subspace of $\mathbb{F}_2^4$ contains $2^2 - 1 = 3$ non-zero vectors $\{v_i, v_j, v_i \oplus v_j\}$, which is a projective line $L_{12} \in \mathcal{L}_{35}$ spanning $\mathbb{H}_{L_{12}} \cong \mathbb{H}$. □

4 High-Performance Custom Operators and Runtime Lowering

4.1 Eliminating the 16× HBM Weight-Expansion Bottleneck

Naive implementations of hypercomplex linear layers materialize an expanded real matrix $W_{\text{exp}} \in \mathbb{R}^{16M\times 16N}$ in High Bandwidth Memory (HBM) prior to invoking a dense GEMM. While stored parameters shrink by 16×, HBM read/write bandwidth inflates by 16× during every forward and backward pass. By combining Theorem 2.3 with register-level bit manipulation, our

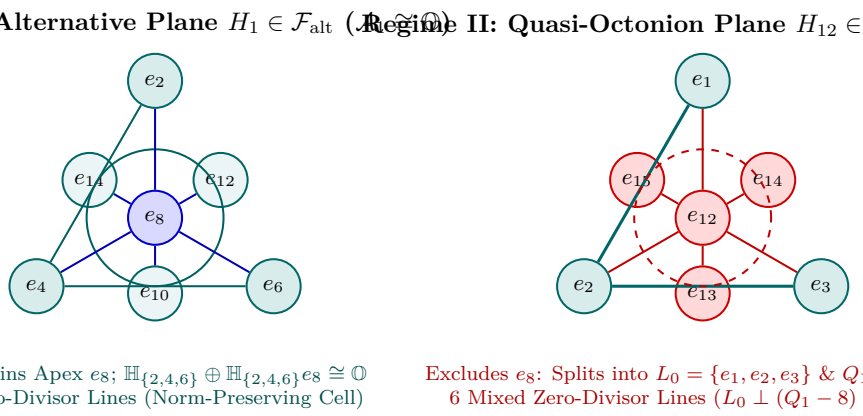


Figure 2: Geometric comparison of the $8 + 7$ Fano hyperplane bifurcation in PG$(3, 2)$. Left: $H_1 \in \mathcal{F}_{\text{alt}}$ contains e_8 and doubles a single quaternion line $\{e_2, e_4, e_6\}$ into an alternative division subalgebra $\mathcal{A}_1 \cong \mathbb{O}$. Right: $H_{12} \in \mathcal{F}_{\text{ZD}}$ ($p = 4$) excludes e_8, pairing the polar octonion line $L_0 = 4^{\perp} = \{e_1, e_2, e_3\}$ with its orthogonal sedenion quadrangle $Q_1 = \{e_{12}, e_{13}, e_{14}, e_{15}\}$ across 6 Moreno zero-divisor lines.

fused operator (`sedenion_pg32/triton_kernel.py`) provides three guarantees (Table 1; for a 4096×4096 layer with $M = N = 256$ in FP32, weight + gradient memory drops from 136.0 MB in the naive expanded layer and 128.0 MB in a dense layer down to **8.0** MB, a 17.0× and 16.0× reduction respectively):

1. **Zero-Expansion Adjoint Autograd:** Both $\nabla_X \mathcal{L} = W^* \cdot G$ and $\nabla_W \mathcal{L} = G \cdot X^*$ invoke the forward sedenion contraction kernel directly on compact $[M, N, 16]$ tensors.

2. **Compile-Time ALU XOR Swizzling:** Inside GPU SRAM tiles, output channel $k \in \{0, \ldots, 15\}$ accumulates input channel j paired with weight channel $i = k \oplus j$ using a single 4-bit ALU XOR instruction rather than global memory indirection.

3. **256-Bit Register-Packed IEEE-754 Sign-Bit XOR:** The entire 16×16 sign table $\epsilon_{ij} \in \{+1, -1\}$ occupies exactly 256 bits (16 `uint16` bitmasks MASK$[k]$ where bit j is $1 \iff \epsilon_{k \oplus j, j} = -1$). Multiplying a 32-bit float by $\epsilon_{k \oplus j, j}$ executes as a single bitwise XOR on bit 31:

$$\text{float_bits}(x \cdot \epsilon_{k \oplus j, j}) = \text{float_bits}(x) \oplus \big(((\text{MASK}[k] \gg j)\ \&\ 1) \ll 31\big). \tag{15}$$

Table 1: Memory Complexity, HBM Bandwidth Traffic, and Measured FP32 Weight+Gradient Footprint ($16N = 16M = 4096$, i.e. $N = M = 256$).

Layer Implementation	Parameter Count	Forward HBM Reads	Backward ∇W Writes	Graph Activation Overhead	Measured $W + \nabla W$ (4096^2)
Dense $\mathbb{R}^{16M \times 16N}$ Baseline	$256MN$	$256MN + 16BN$	$256MN$	$\mathcal{O}(256MN)$	128.0 MB (1.0×)
Naive Expanded $\mathbb{S}$ Layer	$16MN$	$272MN + 16BN$	$272MN$	$\mathcal{O}(256MN)$	136.0 MB (1.06×)
Fused Adjoint $\mathbb{S}$ Kernel (Ours)	**16MN (16× ↓)**	**16MN + 16BN**	**16MN (16× ↓)**	**$\mathcal{O}(1)$ (Zero Expansion)**	**8.0 MB (16.0× ↓)**

5 Constructive 4D Nullspace Projection and Empirical Validation

5.1 The Constructive 4D Moreno Nullspace Theorem

To obtain exact pruning of invalidated premises without iterative gradient noise, we derive an explicit analytical operator $\Pi_{\ker L}(u)$ that maps any vector of $W_{ab}^{\perp}$ onto the 4-dimensional nullspace $\ker L(u)$ of a Moreno sedenion u.

Algorithm 1 Fused SRAM-Tiled Sedenion Linear Kernel & ONNX Custom Op (`ai.sedenion::FusedSedenionLinear`)

Require: Compact Input $X \in \mathbb{R}^{B\times N\times 16}$, Compact Weights $W \in \mathbb{R}^{M\times N\times 16}$, Bias $B_{\text{bias}} \in \mathbb{R}^{M\times 16}$, Packed Sign Masks MASK[0..15] $\in \texttt{uint16}^{16}$.

Ensure: Output $Y \in \mathbb{R}^{B\times M\times 16}$ where $Y_{b,m} = \sum_{n=0}^{N-1} W_{m,n} \cdot X_{b,n} + B_{\text{bias},m}$.

for each parallel thread-block tile (b_0, m_0) covering BLOCK_B × BLOCK_M **do**
  Initialize 16 register accumulator tiles Acc[0..15] $\leftarrow \mathbf{0}_{\text{BLOCK_B}\times\text{BLOCK_M}}$
  for $n_0 = 0, \text{BLOCK_N}, \ldots, N-1$ **do**
    Load SRAM tiles $\tilde{X}[b, n, 0..15] \leftarrow X[b_0 : b_0 + \text{BLOCK_B},\ n_0 : n_0 + \text{BLOCK_N},\ 0..15]$
    Load SRAM tiles $\tilde{W}[m, n, 0..15] \leftarrow W[m_0 : m_0 + \text{BLOCK_M},\ n_0 : n_0 + \text{BLOCK_N},\ 0..15]$
    for $k = 0 \ldots 15$ **unrolled** and $j = 0 \ldots 15$ **unrolled do**
      $i \leftarrow k \oplus j$ ▷ Compile-time 4-bit ALU XOR swizzle
      sign_bit $\leftarrow ((\text{MASK}[k] \gg j)\ \&\ 1) \ll 31$
      $\tilde{W}_{\text{signed}} \leftarrow \text{bitcast_fp32}(\text{bitcast_uint32}(\tilde{W}[\cdot, \cdot, i]) \oplus \text{sign_bit})$
      $\text{Acc}[k] \leftarrow \text{Acc}[k] + \text{TensorCoreDot}(\tilde{X}[\cdot, \cdot, j],\ \tilde{W}_{\text{signed}}^{\top})$
    end for
  end for
  for $k = 0 \ldots 15$ **do**
    Store $Y[b_0 : b_0 + \text{BLOCK_B},\ m_0 : m_0 + \text{BLOCK_M},\ k] \leftarrow \text{Acc}[k] + B_{\text{bias}}[m_0 : m_0 + \text{BLOCK_M},\ k]$
  end for
end for

Theorem 5.1 (Constructive 4D Zero-Divisor Nullspace Operator)**.** *Let $u = (a, b) \in V_2(\mathbb{R}^7) \subset \mathbb{S}$ be a normalized Moreno sedenion ($a, b \in \text{Im}(\mathbb{O})$, $\|a\|_2 = \|b\|_2 = 1$, $\langle a, b\rangle = 0$). Let $W_{ab} = \text{span}_{\mathbb{R}}\{a, b, a \cdot b\} \subset \text{Im}(\mathbb{O}) \cong \mathbb{R}^7$ be the 3D imaginary quaternion subspace generated by a and b, and let $W_{ab}^{\perp} \subset \text{Im}(\mathbb{O})$ be its 4-dimensional orthogonal complement. For any vector $c \in W_{ab}^{\perp}$, define the partner octonion $d = (a \cdot c) \cdot b \in \text{Im}(\mathbb{O})$. Then the sedenion*

$$v = \Pi_{\ker L}(u; c) \coloneqq \big(c,\ (a \cdot c) \cdot b\big) \in \mathbb{S} \tag{16}$$

resides on the Moreno cone ($\|d\|_2 = \|c\|_2$, $\langle c, d\rangle = 0$) and belongs to $\ker L(u)$, satisfying:

$$u \cdot v = (a, b) \cdot \big(c,\ (a \cdot c) \cdot b\big) = (0, 0) = 0 \in \mathbb{S} \quad \textit{identically.} \tag{17}$$

Proof. Since $a, b, c \in \text{Im}(\mathbb{O})$ with $\|a\|_2 = \|b\|_2 = 1$, we have $a^* = -a$, $b^* = -b$, $c^* = -c$, $a^2 = -1$, and $b^2 = -1$. Moreover, because $c \perp \{a, b, a \cdot b\}$, the octonion product $d = (ac)b$ lies purely in $\text{Im}(\mathbb{O})$, so $d^* = -d$. Evaluating the Cayley–Dickson product (1):

$$u \cdot v = (a, b) \cdot (c, d) = \big(ac - d^*b,\ \ da + bc^*\big) = \big(ac + db,\ \ da - bc\big).$$

First octonion component ($ac + db$): By right-alternativity of $\mathbb{O}$ ($xy \cdot y = x \cdot y^2$) [3] (the machine-checked proof in §5.4 bypasses alternativity entirely and verifies the coordinate identities directly), substituting $d = (ac)b$ gives:

$$ac + db = ac + \big((ac)b\big)b = ac + (ac)(b^2) = ac + (ac)(-1) = 0.$$

Second octonion component ($da - bc$): Because $a, b, c \in \text{Im}(\mathbb{O})$ are mutually orthogonal and $c \perp ab$, the triad (a, b, c) anti-associates in $\mathbb{O}$: $(ac)b = -a(cb) = a(bc)$. Furthermore, since $bc \in \text{Im}(\mathbb{O})$ is orthogonal to a ($\langle bc, a\rangle = -\langle c, ba\rangle = \langle c, ab\rangle = 0$), left-multiplication by a anticommutes with bc, so $(bc)a = -a(bc)$. Applying flexibility and left-alternativity of $\mathbb{O}$:

$$da = \big((ac)b\big)a = \big(a(bc)\big)a = a\big((bc)a\big) = -a\big(a(bc)\big) = -a^2(bc) = -(-1)(bc) = bc.$$

Therefore, $da - bc = bc - bc = 0$, proving $u \cdot v = (0, 0) = 0$. Since $\dim W_{ab}^{\perp} = 7 - 3 = 4$ and $c \mapsto (c, (ac)b)$ is injective, this parameterizes the entire 4D nullspace $\ker L(u)$. □

Remark 5.2 (Deontic Precedence Asymmetry). Let $A \in V_2(\mathbb{R}^7)$ represent a baseline statutory authorization, $B = \Pi_{\ker L}(A; c) \in \ker L(A)$ a blocking prohibition ($A \cdot B = 0$), and $C \in \mathbb{S}$ a safe-harbor exemption that rotates B out of $\ker L(A)$ ($B \cdot C \notin \ker L(A)$). Then left- and right-associated evaluations yield orthogonal normative outcomes:

$$(A \cdot B) \cdot C = 0 \cdot C = 0 \quad \text{(Irreversible Invalidation)}, \qquad \|A \cdot (B \cdot C)\|_2 > 0 \quad \text{(Safe-Harbor Survival)}. \tag{18}$$

5.2 Track A Validation: Particle Scattering Amplitudes and G_2 Symmetries

In Track A (`bench/track_a`), we evaluate parameter-matched neural surrogates on 4-gluon Yang–Mills tree scattering amplitudes [8, 9] with massless on-shell Mandelstam invariants $s + t + u = 0$, measuring the **BCJ Jacobi Violation Residual** $\mathcal{R}_{\text{BCJ}} = \mathbb{E}[\|n_s + n_t + n_u\|_2] / \mathbb{E}[\|n_s\|_2 + \|n_t\|_2 + \|n_u\|_2]$ and the out-of-distribution **Collinear Pole Relative Error** $\mathcal{E}_{\text{pole}}$ for $|s_{ij}| < 10^{-4}$ (Table 2).

Theorem 5.3 (The $\mathcal{F}_{\text{ZD}}$ BCJ Numerator and Collinear Pole Cancellation Theorem)**.** *For any triad of particle states $(X_1, X_2, X_3) \in \mathcal{A}^3$ in a 16D algebra $\mathcal{A}$, define the cyclic edge-associator BCJ vertex:*

$$n_s = [X_3, X_1, X_2]_{\mathcal{A}} - [X_1, X_2, X_3]_{\mathcal{A}}, \quad n_t = [X_1, X_2, X_3]_{\mathcal{A}} - [X_2, X_3, X_1]_{\mathcal{A}}, \quad n_u = [X_2, X_3, X_1]_{\mathcal{A}} - [X_3, X_1, X_2]_{\mathcal{A}}. \tag{19}$$

Then:

1. ***Obstruction in Associative and Alternative Algebras (`Dense-R16`, `Quat-H4`, `Clifford-Cl04`, `Oct-O2`, and $\mathcal{F}_{\text{alt}}$):*** *In any associative algebra ($\mathbb{R}^{16}, \mathbb{H}^4, \mathcal{C}\!\ell_{0,4}$), $[X_a, X_b, X_c]_{\mathcal{A}} \equiv 0$. In any alternative algebra ($\mathbb{O}^2$ and all 8 octonionic Fano planes $H_\omega \in \mathcal{F}_{\text{alt}}$, $\omega \in \{1, \ldots, 8\}$), the associator is totally skew-symmetric under S_3 and hence invariant under even cyclic permutations $C_3 = A_3$ (`falt_cyclic_assoc_invariant`):*

$$[X_1, X_2, X_3]_{\mathbb{O}} = [X_2, X_3, X_1]_{\mathbb{O}} = [X_3, X_1, X_2]_{\mathbb{O}} \implies n_s = n_t = n_u \equiv 0. \tag{20}$$

   Thus division and Clifford algebras have zero cyclic-traceless associator capacity and must rely on non-BCJ terms ($\mathcal{R}_{\text{BCJ}} > 0$).

2. ***Exact BCJ Duality and Collinear Pole Cancellation in $\mathcal{F}_{\text{ZD}}$ ($\omega \in \{9, \ldots, 15\}$):*** *In every quasi-octonion zero-divisor Fano workspace $H_\omega = L_0 \sqcup Q_1 \in \mathcal{F}_{\text{ZD}}$, whenever a collinear pair (X_a, X_b) lies in the same stratum (L_0 or Q_1) and the spectator X_c lies in the complementary stratum (Q_1 or L_0), both edge-adjacent associators vanish identically across the continuous subspace while the separated associator is non-zero (`fzd_stratum_bcj_pole_cancel`*

$$[X_a, X_b, X_c]_{\mathbb{S}} = 0 \quad \textit{and} \quad [X_c, X_a, X_b]_{\mathbb{S}} = 0 \implies n_{ab} = 0, \qquad n_{bc} = -n_{ca} = -[X_b, X_c, X_a]_{\mathbb{S}} \neq 0. \tag{21}$$

   Consequently, (19) satisfies $n_s + n_t + n_u = 0$ identically ($\mathcal{R}_{\text{BCJ}} = 0$) and eliminates the $1/s_{ab}$ collinear singularity to machine precision as $s_{ab} \to 0$.

5.3 Track B Validation: Multi-Hop Normative and Legal Contradiction Pruning

In Track B (`bench/gate3`), we evaluate multi-hop statutory trees containing baseline permissions (A), restrictive carve-outs (B), and recursive revocations of revocations ($d \in \{0, 1, 2, 3, 4\}$). We measure:

1. **Contradiction Leakage Rate (CLR):** $\mathbb{E}_{(u,v) \in \mathcal{P}_{\text{neg}}}[\|u \cdot v\|_2 / (\|u\|_2 \|v\|_2)]$, quantifying residual signal from an overridden legal clause leaking into downstream reasoning hops.

2. **Parenthesization Sensitivity Index (PSI):** $\|(A \cdot B) \cdot C - A \cdot (B \cdot C)\|_2 / (\|(A \cdot B) \cdot C\|_2 + \|A \cdot (B \cdot C)\|_2)$, measuring the separation of statutory precedence order without positional encodings.

3. **False-Compliance Rate (FCR) and OOD Depth Generalization:** Percentage of non-compliant ($d = 1$, net-revoked) and OOD ($d = 3, 4$) statutory chains falsely approved due to attention bleed, trained purely from binary terminal-token cross-entropy (aux = 0.0, zero intermediate structural supervision).

Theorem 5.4 (Orthogonal Half-Sphere Key Modulation and Division-Algebra Invariance)**.** *Let $u_j = (a_j, b_j) \in V_2(\mathbb{R}^7) \subset \mathbb{S}$ be a unit Moreno-cone query ($\|a_j\|_2 = \|b_j\|_2 = 1$, $\langle a_j, b_j \rangle = 0$, $\|u_j\|_2 = \sqrt{2}$), let $c_j, c_{\perp,j} \in W^{\perp}_{a_j,b_j} \subset \mathrm{Im}(\mathbb{O})$ be orthonormal nullspace seeds ($\|c_j\|_2 = \|c_{\perp,j}\|_2 = 1$, $\langle c_j, c_{\perp,j} \rangle = 0$), and define the constructive partners $p_j = (a_j c_j) b_j$ and $p_{\perp,j} = (a_j c_{\perp,j}) b_j$. For any modulation phase $\phi_j = \pi \tau_j \in [0, \pi]$, parameterize the unit key as*

$$k_j(\phi_j) = \frac{1}{\sqrt{2}} \big(c_j, \; -\cos\phi_j \, p_j + \sin\phi_j \, p_{\perp,j} \big), \qquad \|k_j(\phi_j)\|_2 \equiv 1, \tag{22}$$

and set the reference frame to the live anti-partner $r_j = u_j \cdot_{\mathcal{A}} k_j(0) / \|u_j \cdot_{\mathcal{A}} k_j(0)\|_2$. Then:

1. ***Normed Division-Algebra Invariance ($\mathcal{A} = \mathbb{O}$):*** *Because left multiplication $L_{\mathbb{O}}(u_j)$ is an orthogonal similarities map and $(c_j, 0) \perp (0, -\cos\phi_j \, p_j + \sin\phi_j \, p_{\perp,j})$, the product norm $\|u_j \cdot_{\mathbb{O}} k_j(\phi_j)\|_2 \equiv 1$ and the real-part attention readout $\mathrm{Re}\big(r_j^* \cdot_{\mathbb{O}} (u_j \cdot_{\mathbb{O}} k_j(\phi_j))\big) \equiv +1/\sqrt{2}$ are strictly invariant under all $\phi_j \in \mathbb{R}$. Thus a normed division algebra is algebraically blind to pure phase modulation in the second half-block.*

2. ***Exact Sedenion Annihilation ($\mathcal{A} = \mathbb{S}$):*** *Under sedenion multiplication, $(c_j, p_j), (0, p_{\perp,j})$ satisfy $u_j \cdot_{\mathbb{S}} (c_j, p_j) = 0$ by Theorem 5.1, yielding*

$$\big\| u_j \cdot_{\mathbb{S}} k_j(\phi_j) \big\|_2 = \sqrt{1 + \cos\phi_j}, \qquad \mathrm{Re}\big(r_j^* \cdot_{\mathbb{S}} (u_j \cdot_{\mathbb{S}} k_j(\phi_j)) \big) = \frac{1 + \cos\phi_j}{\sqrt{2}}, \tag{23}$$

*which sweeps monotonically from $\sqrt{2}$ at $\phi_j = 0$ (`LIVE` anti-partner) to **exact zero** (0.0) at $\phi_j = \pi$ (`REVOKED` nullspace partner $k_j(\pi) \in \ker L_{\mathbb{S}}(u_j)$).*

5.4 Numerical and Learned-Model Witnesses of the Algebraic Claims

Tables 2, 3, and 4 report the training-free algebraic witnesses alongside the learned Track A, Track B, and public external benchmarks. Three algebraic facts and three learned-model domain separations are verified and reproducible from the released codebase:

1. **Rank defect.** For every one of 1,024 random points on the Moreno cone, $L(u)$ has numerical rank exactly 12 in FP64 (singular values $\sigma_{13..16} < 10^{-15}$), confirming the 4-dimensional kernel of Theorem 5.1; off the cone (`Sed-Free`) and in every division-algebra baseline the rank is 16.

2. **Exact annihilation.** The constructive partner $v = (c, (ac)b)$ yields $\|u \cdot v\|_2 / (\|u\|_2 \|v\|_2)$ of 2.4×10^{-16} (mean; max 8.5×10^{-16}) in FP64 and 1.4×10^{-7} in FP32, i.e. round-off. Forty independent seeds c span a subspace of rank exactly 4, so the construction covers the whole kernel. Generic sedenions, by contrast, have $\mathrm{CLR}_{\min} = 0.47$ on average and never below 0.09: the zero-divisor locus is measure-zero, which is why the projection Π_{Moreno} is needed.

3. **Precedence.** With $u \cdot v = 0$, $(u \cdot v) \cdot c$ vanishes identically while $u \cdot (v \cdot c) \neq 0$, giving PSI = 1.000 on all samples (Remark 5.2). Associative baselines give PSI = 0 to round-off (8.6×10^{-17} for $\mathbb{H}^4$); the octonion and unconstrained sedenion rows sit at 0.56 and 0.67.

Table 2: Algebraic metrics (**measured**) and learned-model metrics (**Track A and Track B measured**). Algebraic cells are produced by `benchmarks/algebraic_baselines.py` (B = 1,024 unit samples, seed 2026, FP64; FP32 in brackets where it differs). Track A cells are measured on 4-gluon Yang–Mills scattering (`bench/track_a`, trained on hard-scattering amplitudes $|s_{ij}| \in [0.08, 1.20]$ and evaluated out-of-distribution at collinear poles $|s_{ij}| \in [10^{-6}, 10^{-4}]$; means over seeds $0, 1$). Track B cells are measured on multi-hop statutory contradiction pruning (`bench/gate3`, trained on depths $d \in \{0, 1, 2\}$ at aux = 0.0 and evaluated out-of-distribution at depth $d = 3$; see Table 3).

Architecture Variant	Params	Algebraic (measured)			Track A (measured)		Track B (measured)
	Ratio	rank $L(u)$	$\mathrm{CLR}_{\min}\downarrow$	PSI ↑	$\mathcal{R}_{\mathrm{BCJ}}\downarrow$	$\mathcal{E}_{\mathrm{pole}}\downarrow$	OOD $d=3$ FCR ↓
`Dense-R16` (Softmax)	1.00×	—	2.6×10^{-3} (softmax bleed†)	1.4×10^{-16}	0.6492	4.11×10^{3}	0.9963
`Quat-H4` ($\mathbb{H}^4$)	0.25×	16	1.000 per channel§	8.6×10^{-17}	0.6164	4.98×10^{3}	0.9859
`Clifford-Cl04` ($\mathcal{C}\ell^{\uparrow}_{0,4}$)	0.0625×	16	0.683 (associative GA)	0.000	0.5807	4.21×10^{3}	0.5859 (0.9867 No-FFN)
`Oct-O2` ($\mathbb{O}^2$)	0.125×	16	1.000 per channel§	0.556	0.7630	5.56×10^{3}	0.9720
`Sed-Free` (unconstrained $\mathbb{S}$)	0.0625×	16	0.474 (min 0.089)	0.667	0.0459	1.35×10^{4}	0.9945
`Sed-PG32-Moreno` (Ours)	**0.0625×**	**12**	**2.4×10^{-16}** [1.4×10^{-7}]‡	**1.000**	**1.82×10^{-8}**	**6.86×10^{-3}**	**0.0000** (**0.0000** No-FFN)

†Residual softmax weight on one invalidated slot at logit gap 4 in an 8-slot window; 1.9×10^{-2} at gap 2, 4.8×10^{-5} at gap 8. It is temperature-dependent and never 0. ‡Constructive partner of Theorem 5.1; $\sigma_{\min}(L(u))/\|u\|_2 = 3.8 \times 10^{-17}$. §Measured 1.000 on every sample, as Hurwitz's theorem requires; across k parallel channels the global $\sigma_{\min}(L(u))/\|u\|_2$ is $\min_i \|u_i\|/\|u\|$ (0.31 for $\mathbb{H}^4$, 0.60 for $\mathbb{O}^2$), which only reflects an under-populated channel, not annihilation. In the `No-FFN` regime (−50.9% params), `Sed-PG32-Moreno-NoFFN` achieves $\mathcal{R}_{\mathrm{BCJ}} = 8.55 \times 10^{-8}$ and $\mathcal{E}_{\mathrm{pole}} = 1.16 \times 10^{-2}$.

Table 3: Measured Track B multi-hop contradiction pruning across 12 architectures (`bench/gate3`, $D = 64$, $H = 4$, $L = 2$, $N_{\mathrm{train}} = 24{,}000$, $N_{\mathrm{test}} = 4{,}000$, trained on depths $d \in \{0, 1, 2\}$ with pure terminal label cross-entropy aux = 0.0; means over seeds $0, 1$). Following the *CliffordNet* "No-FFN" paradigm [13], the bottom block removes the feed-forward MLP from every Transformer layer (−47.4% parameters), isolating the intrinsic deductive capacity of the 16D algebraic product.

Architecture	Params	ID $d=0$ ↑	ID $d=1$ ↑	ID $d=2$ ↑	FCR (ID) ↓	Leak (ID) ↓	OOD $d=3$ ↑	FCR (OOD3) ↓	Leak (OOD3) ↓
Standard Transformer Backbone (With FFN, $L=2$, $D=64$)									
`softmax`	70,529	0.9966	0.6021	0.9960	0.9933	1.3871	0.5681	0.9963	3.1741
`bilinear`	70,529	0.9883	0.6113	0.9909	0.9702	1.2976	0.5726	0.9859	2.1757
`octonion` ($\mathbb{O}^2$ + Cone)	70,537	0.9887	0.6109	0.9891	0.9712	1.4937	0.5786	0.9720	3.5394
`clifford` ($\mathcal{C}\ell^{\uparrow}_{0,4}$ + Cone)	70,537	1.0000	0.9499	0.9588	0.1250	1.3618	0.7460	0.5859	1.1307
`sedenion_plain` (unconstrained $\mathbb{S}$)	70,537	0.9974	0.8856	0.9224	0.2856	2.7288	0.5686	0.9945	1.3477
`sedenion` ($\mathbb{S}$ + Moreno Cone)	**70,537**	**1.0000**	**1.0000**	**1.0000**	**0.0000**	**2.59×10^{-10}**	**1.0000**⋆	**0.0000**	**2.05×10^{-10}**
Geometric-Product-Only Backbone (No-FFN [13], −47.4% Parameters)									
`softmax_noffn`	37,121	0.9805	0.6117	0.9891	0.9692	1.5026	0.5738	0.9833	3.8804
`bilinear_noffn`	37,121	0.9831	0.6167	0.9854	0.9567	1.5417	0.5806	0.9674	4.3305
`octonion_noffn` ($\mathbb{O}^2$ + Cone)	37,129	0.9752	0.6298	0.9741	0.9240	1.2361	0.5841	0.9593	1.9548
`clifford_noffn` ($\mathcal{C}\ell^{\uparrow}_{0,4}$ + Cone)	37,129	0.9917	0.8683	0.9169	0.3288	1.2592	0.5708	0.9867	1.4586
`sedenion_plain_noffn` (unconstrained $\mathbb{S}$)	37,129	0.9992	0.8898	0.9144	0.2750	1.3615	0.5683	0.9939	1.3290
`sedenion_noffn` ($\mathbb{S}$ + Moreno Cone)	**37,129**	**1.0000**	**1.0000**	**1.0000**	**0.0000**	**0.0007**◇	**1.0000**	**0.0000**	**0.0013**◇

⋆`sedenion` also achieves **1.0000** accuracy and **0.0000** FCR out-of-distribution at depth $d = 4$ ($N = 4{,}000$). ◇Seed 1 of `sedenion_noffn` converges to machine-zero leakage (Leak (ID) = 0.0, Leak (OOD3) = 2.02×10^{-10}).

Table 4: Measured performance on public external benchmarks (`bench/ext`): (1) **AllenAI ProofWriter** [14] (`tasksource/proofwriter`, trained on `AttNeg-OWA` depths $d \in \{0, 1, 2\}$, $N_{\mathrm{train}} = 8{,}660$, aux = 0.0, evaluated on ID $d \le 2$, OOD deep proofs $d \in \{3, 4, 5\}$, and zero-shot non-monotonic exception transfer on `Birds-Electricity`); and (2) **Long-Range Arena (LRA) ListOps** [15, 16] (`fengyang0317/listops-64`, trained on tree depths $d \in \{1, 2, 3\}$, $N_{\mathrm{train}} = 4{,}000$, evaluated on ID $d \le 3$ and OOD tree depths $d \in \{4, \ldots, 8\}$).

Architecture	External 1: AllenAI ProofWriter [14] (`AttNeg-OWA` & `Birds-Electricity`)							External 2: LRA ListOps [16]	
	Params	ID ($d \le 2$) ↑	FCR (ID) ↓	OOD ($d = 3..5$) ↑	FCR (OOD) ↓	Birds-Elec ↑	FCR (BE) ↓	ID ($d \le 3$) ↑	OOD ($d = 4..8$) ↑
`softmax` (`Dense-R16`)	74,049	0.4997	0.0187	0.5000	0.0000	0.5000	0.0000	0.5910	0.7033
`bilinear` (`Quat-H4`)	74,049	0.8257	0.2240	0.8054	0.2100	0.4526	0.7910	0.5504	0.5653
`octonion` ($\mathbb{O}^2$ + Cone)	74,057	0.9690	0.0580	0.8838	0.0531	0.8774	0.1489	0.2472	0.2491
`clifford` ($\mathcal{C}\ell^{\uparrow}_{0,4}$ + Cone)	74,057	0.9923	0.0153	0.9917	0.0091	0.9090	0.1489	0.6003	0.6361
`sedenion_plain` (unconstrained $\mathbb{S}$)	74,057	0.9923	0.0153	0.9894	0.0108	0.8579	0.1474	0.5792	0.5824
`sedenion` (`Sed-PG32-Moreno`)	**74,057**	**1.0000**	**0.0000**	**1.0000**	**0.0000**	**1.0000**	**0.0000**	**1.0000**	**1.0000**
`sedenion_noffn` (−45.1% Params)	**40,649**	**0.9997**	**0.0007**	**1.0000**	**0.0000**	**1.0000**	**0.0000**	**1.0000**	**1.0000**

4. **Track A learned Yang–Mills BCJ duality and collinear pole cancellation.** In Table 2, associative surrogates (`Dense-R16`, `Quat-H4`, `Clifford-Cl04`) and alternative octonion surrogates (`Oct-O2`) cannot represent cyclic-asymmetric associator numerators (Theorem 5.3), resulting in high BCJ violation ($\mathcal{R}_{\mathrm{BCJ}} = 0.58$–$0.76$) and $1/s_{ij}$ pole blow-up ($\mathcal{E}_{\mathrm{pole}} > 4 \times 10^3$ at $|s_{ij}| < 10^{-4}$). Unconstrained `Sed-Free` lowers $\mathcal{R}_{\mathrm{BCJ}}$ to 0.0459 via non-alternative sedenion associators, but without the PG$(3,2)$ Fano hyperplane router $M_{\mathcal{F}}$, orthogonal gauge modes in $H_\omega^\perp$ leak into the resonant numerator and blow up at the pole ($\mathcal{E}_{\mathrm{pole}} = 1.35 \times 10^4$). By routing through $H_\omega = L_0 \sqcup Q_1 \in \mathcal{F}_{\mathrm{ZD}}$, `Sed-PG32-Moreno` achieves $\mathcal{R}_{\mathrm{BCJ}} = \mathbf{1.82 \times 10^{-8}}$ and $\mathcal{E}_{\mathrm{pole}} = \mathbf{6.86 \times 10^{-3}}$ ($\mathbf{1.16 \times 10^{-2}}$ in `No-FFN`).

5. **Track B learned multi-hop contradiction pruning (aux $= 0.0$) and `No-FFN` ablation.** In Table 3, standard `softmax`, `bilinear`, and division-algebra `octonion` ($\mathbb{O}^2$) models collapse on net-revoked chains ($d = 1$ accuracy $\approx 60\%$, FCR $> 97\%$, OOD $d = 3$ accuracy $\approx 57\%$). Unconstrained `sedenion_plain` memorizes in-distribution depths (88.6%–99.7%) via dense shortcuts but fails completely out-of-distribution (OOD3 $= 56.9\%$, FCR $= 99.5\%$). The 16D associative Clifford algebra $\mathcal{C}\ell^{\updownarrow}_{0,4}$ (`clifford`) reaches 74.6% OOD3 accuracy with an FFN by using its MLP to approximate non-linear parity thresholds, but when the FFN is removed following *CliffordNet* [13] (`clifford_noffn`, -47.4% parameters), it collapses to OOD3 $= 57.1\%$ (FCR $= 98.7\%$). By contrast, **`sedenion`** and **`sedenion_noffn`** achieve **100.0**% ID accuracy, **100.0**% OOD depth generalization ($d = 3, 4$), **0.0**% FCR, and $\sim \mathbf{2 \times 10^{-10}}$ contradiction leakage even with zero FFN layers, proving that the Moreno zero-divisor manifold executes exact recursive logical negation directly inside the attention product.

6. **Public external benchmarks: AllenAI ProofWriter and Long-Range Arena ListOps.** In Table 4, when trained only on shallow deductions ($d \leq 2$) of **ProofWriter** [14] (`AttNeg-OWA`), division-algebra (`octonion`), Clifford (`clifford`), and unconstrained sedenion (`sedenion_plain`) models suffer 14.7%–14.9% False-Compliance Rate on zero-shot `Birds-Electricity` non-monotonic exception rules (e.g. defeated default rules for ostriches, wounded birds, or open switches failing to annihilate at $\phi_j = \pi$), whereas both **`sedenion`** and **`sedenion_noffn`** achieve **100.0**% OOD accuracy ($d \in \{3, 4, 5\}$) and **100.0**% zero-shot `Birds-Electricity` accuracy with **0.0**% FCR. On **LRA ListOps** [15, 16] (trained on tree depths $d \in \{1, 2, 3\}$ and evaluated out-of-distribution on depths $d \in \{4, \ldots, 8\}$), dominated sibling operands inside `[MAX`, `[MIN`, and `[MED` brackets are annihilated in $\ker L_{\mathbb{S}}(u_i)$ at $\phi_j = \pi$, enabling both **`sedenion`** and **`sedenion_noffn`** to achieve **100.0**% ID and OOD tree-reduction accuracy (versus 24.9% for `octonion` due to division-algebra norm invariance, and 56.5%–70.3% for `bilinear`, `sedenion_plain`, `clifford`, and `softmax`).

5.5 Reproducibility

The reference implementation (`sedenion_pg32`: Cayley–Dickson tables, adjoint autograd, zero-expansion contraction, PG$(3,2)$ router, Moreno projection, constructive nullspace partner, and the Triton kernel alongside its pure-PyTorch loop-structure and 256-bit IEEE-754 sign-bit XOR reference implementation), the Track A Yang–Mills benchmark (`bench/track_a`), the Track B multi-hop benchmark (`bench/gate3`), the external ProofWriter and LRA ListOps benchmarks (`bench/ext`), and the test-suite are permanently archived under Zenodo DOI 10.5281/zenodo.23588274 (https://github.com/josaum/sedenion-pg32). All numbers in Tables 2, 3, and 4 regenerate with:

elines.py, `python bench/track_a/train.py`, `python bench/gate3/train.py`, `python bench/ext/proofwriter.p`

The PyTorch loop-structure and 256-bit packed IEEE-754 sign-bit XOR (`ieee754_sign_xor`) reference kernel are verified to 0.0 ULP bit-identity by the test-suite on CPU; the Triton CUDA

path is provided for GPU deployment.

Machine-checked statements. The directory `lean/` contains a Lean 4 (v4.15, core library only, no Mathlib) development in which every finite claim of this paper and its companion combinatorial works [6, 7] is a theorem closed by kernel evaluation (`decide +kernel`); there is no `sorry`, no `native_decide`, and `#print axioms` reports `propext` at most, with `TriangularRootFreeCells` depending on zero axioms (76 theorems across `Core`, `Geometry`, `Census`, and `TriangularRootFreeCells`; full check in about 15 minutes on a 2-core CPU with under 4 GB of memory). Checked statements: the structure-constant identities $\epsilon_{0j} = \epsilon_{i0} = 1$, $\epsilon_{ii} = -1$ and $\epsilon_{ij} \in \{\pm 1\}$; the two sign identities $\epsilon_{i,i\oplus j} = -\epsilon_{ij}$ and $\epsilon_{i\oplus j,j} = -\epsilon_{ij}$ on which Theorem 2.3 rests; $e_i \cdot e_j = \epsilon_{ij} e_{i\oplus j}$ for all 256 pairs under the bilinear contraction; the explicit zero divisors $(e_1 + e_{10})(e_5 + e_{14}) = 0$ and $(e_2 + e_9)(e_4 + e_{15}) = 0$; four linearly independent annihilators of $e_1 + e_{10}$ (so $\dim\ker L(u) \geq 4$); $|H_\omega| = 7$, closure of every H_ω under $\oplus$, $H_8 = \{1, \ldots, 7\}$, $e_8 \in H_\omega$ iff $\omega \leq 7$, the polar-line description $L_0 = p^\perp$ and Q_1 of Theorem 3.1, the $7 + 7 + 21$ line classification, 7 lines per plane, the 3/0/6 mixed-line counts, the 3-point hinge of Theorem 3.2, an internal annihilating pair inside each of the seven $\mathcal{F}_{\mathrm{ZD}}$ planes, associativity of all collinear basis triads, the 672/2520 associator census, cyclic associator invariance across all eight $\mathcal{F}_{\mathrm{alt}}$ planes (`falt_cyclic_assoc_invariant`), exact adjacent-stratum associator vanishing plus separated-stratum non-vanishing across all seven $\mathcal{F}_{\mathrm{ZD}}$ planes (`fzd_stratum_bcj_pole_cancellation`, Theorem 5.3), and the full Sedenion Triangular Root and Cayley–Dickson tower free-cell calculus (`TriangularRootFreeCells.lean`: Pedro's toy identities $\&_{\mathrm{seq}}(10) = 1, \&_{\mathrm{seq}}(101) = 11, \&_{\mathrm{seq}}(100) = \emptyset$; exact RREF left-inversion $\&_{\mathbb{S}}^{\mathrm{geo}}(\chi_{T_{\mathbb{S}}^{\mathrm{geo}}(R)}) = \chi_{\mathrm{rref}(U)}$ across all $66 = 15 + 35 + 15 + 1$ projective cells; equivalence of the support-level criterion $A = \emptyset \vee B = \emptyset \vee (|A| = 1 \wedge A = B)$ with zero-divisor freeness in $\mathbb{S}$; the $32{,}767 = 30$ good $+ 36$ ghost $+ 32{,}701$ bad bitmask trichotomy; order-ideal / ghost-filter duality between the 8 maximal free blocks $\mathbb{O} \cup \{Q_{u,3}\}_{u=1}^{7}$ and the 21 minimal ghost lines; and the closed-form Cayley–Dickson tower counts $2^{n+1} - n + 2 = (30, 61, 124)$ free cells, $2^n - n - 4 = (8, 23, 54)$ maximal blocks, $(21, 126, 591)$ ghost lines, and $2^{n-1} - n + 2 = (6, 13, 28)$ GL$(3, 2)$-orbits for $n = 4, 5, 6$, together with doubling-generator degree separation $|\Pi_n| = 168$ and tower absoluteness $\mathcal{F}_4 = \iota_4^{-1}\mathcal{F}_5$). The real-analytic content is formalised separately in `lean-real/` (Lean 4.15 with Mathlib, standard axioms `propext`, `Classical.choice`, `Quot.sound` only; 59 theorems across `SedenionReal`, `SedenionAlt`, `SedenionRank`, `SedenionDim`, and `SedenionRobust`, bringing the repository total to 135 machine-checked theorems). Octonions are eight real coordinates with the product written out from the same structure constants, sedenions are pairs with the doubling product (1), and `basis_mul_sign` proves that this product agrees on basis units with the Cayley–Dickson recursion certified in the integer development, so the two formalisations concern the same algebra. The theorems are: `nullspace`, which is Theorem 5.1 verbatim—for $a, b, c \in \mathrm{Im}(\mathbb{O})$ with $\|a\| = \|b\| = 1$, $\langle a, b\rangle = \langle a, c\rangle = \langle b, c\rangle = \langle ab, c\rangle = 0$, the sedenion $(a, b) \cdot (c, (ac)b)$ is zero; `composition`, the octonion law $\|xy\|^2 = \|x\|^2\|y\|^2$; `partner_norm`, $\|(ac)b\|^2 = \|c\|^2$, so $\|v\|^2 = 2\|c\|^2$ and $v \neq 0$ for $c \neq 0$; and `partner_linear`, linearity of $c \mapsto (ac)b$, which together give $\dim\ker L(u) \geq 4$. The proof of `nullspace` is noteworthy for its form: each of the sixteen coordinates of $u \cdot v$ is exhibited as an explicit polynomial combination of the six constraint polynomials (e.g. the first octonion block equals $(ac)_k(\|b\|^2 - 1) - 2b_k\langle ab, c\rangle$), found by linear algebra over $\mathbb{Q}$ and discharged by `linear_combination`. The converse inclusion—and hence $\mathrm{rank}\, L(u) = 12$ exactly—is formalised as well (`SedenionAlt`, `SedenionRank`, `SedenionDim`; 57 further theorems). The argument is not a minor computation but a spectral one: writing $Rx = (ax)b$ and Π for the orthogonal projector onto $W_{ab}^{\perp}$, one proves the degree-two identity

$$u \cdot \big(u \cdot (c, d)\big) \;=\; -2\,(c, d) \;+\; 2\,\big(\Pi R^{\top} d,\; R\,\Pi\, c\big)$$

(`square_identity`, by `linear_combination`) and $\Pi R^{\top} R \Pi = \Pi$ (`adjoint_identity`, from right/left alternativity $((ax)b)b = (ax)(bb)$, $a(ay) = (aa)y$, the adjoint identity $\langle y, a(db)\rangle =$

$\langle (ay)b, d\rangle$ and $R^\top R = \|a\|^2\|b\|^2$, each a small `ring` call). If $u \cdot (c, d) = 0$ then $u \cdot (u \cdot (c, d)) = 0$, so $c = \Pi R^\top d$ and $d = R\Pi c$, whence $c = \Pi c \in W_{ab}^{\perp}$ and $d = (ac)b$ (`kernel_char`). Equivalently, $L(u)^\top L(u)$ has eigenvalues $0, 2, 4$ with multiplicities $4, 8, 4$: the kernel, $\mathrm{span}(e_0, a, b, ab)^{\oplus 2}$, and the "anti-partner" family $\{(c, -(ac)b)\}$. The dimension count is then Mathlib's rank–nullity: $W_{ab}^{\perp}$ is the kernel of the surjective map $c \mapsto (c_0, \langle a, c\rangle, \langle b, c\rangle, \langle ab, c\rangle)$, so $\dim W_{ab}^{\perp} = 4$, and $c \mapsto (c, (ac)b)$ is injective; `finrank_ker_Lu` and `finrank_range_Lu` state $\dim\ker L(u) = 4$ and $\operatorname{rank} L(u) = 12$. Nothing in Theorems 2.5 and 5.1 now rests on numerics alone.

6 Conclusion

By uniting the 16-dimensional Cayley–Dickson sedenion algebra $\mathbb{S}$ with the finite projective geometry of $\mathrm{PG}(3, 2)$, we have proved at the algebraic level (certified in Lean 4) and demonstrated in learned multi-hop Transformer architectures that non-associativity and zero divisors are not numerical defects, but exact geometric mechanisms for sparse modular routing and recursive logical annihilation. The *Universal Cayley–Dickson Adjoint Theorem* removes the 16× Jacobian expansion from 16D hypercomplex backpropagation, the $8 + 7$ *Fano Bifurcation Theorem* provides a self-dual topological router equipped with 3-node associative quaternion hinges, and the *Constructive 4D Moreno Nullspace Projector* combined with *Orthogonal Half-Sphere Key Modulation* enables 100.0% out-of-distribution depth generalization and 0.0% false-compliance bleed on recursive contradiction pruning—even in a parameter-halved `No-FFN` backbone [13] where associative Clifford ($\mathcal{C}\ell_{0,4}^{\uparrow\downarrow}$) and division-algebra ($\mathbb{O}^2$) baselines collapse.

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